

Q1.)

Convert this binary number to its decimal equivalent

a) $(1011.101)_2$

$(1011)_2$

$$\begin{array}{r}
 \times \begin{array}{cccc} 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 0 & 1 & 1 \end{array} \\
 \hline
 8 + 0 + 2 + 1 \\
 = 11
 \end{array}$$

$(.101)_2$

$$\begin{array}{r}
 \times \begin{array}{ccc} 2^{-1} & 2^{-2} & 2^{-3} \\ 1 & 0 & 1 \end{array} \\
 \hline
 \frac{1}{2} \times 1 + \frac{1}{4} \times 0 + \frac{1}{8} \times 1 \\
 0.5 + 0 + 0.125 = 0.625
 \end{array}$$

$$(1011.101)_2 = (11.625)_{10}$$

b) $(1001.011)_2$

$(1001)_2 = (9)_{10}$

$$\begin{array}{r}
 \times \begin{array}{cccc} 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 0 & 0 & 1 \end{array} \\
 \hline
 8 + 0 + 0 + 1 = 9
 \end{array}$$

$(.011)_2 = (.375)_{10}$

$$\begin{array}{r}
 \times \begin{array}{ccc} 2^{-1} & 2^{-2} & 2^{-3} \\ 0 & 1 & 1 \end{array} \\
 \hline
 0 + \frac{1}{4} + \frac{1}{8} \\
 0.25 + 0.125 \\
 = 0.375
 \end{array}$$

$$(1001.011)_2 = (9.375)_{10}$$

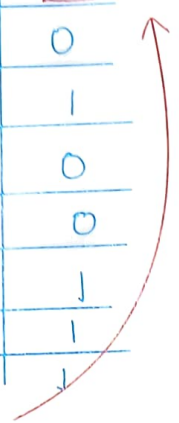
c) $(1101001011)_2$

$$\begin{array}{r}
 \times \begin{array}{cccccccccccc} 2^9 & 2^8 & 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{array} \\
 \hline
 512 + 256 + 0 + 64 + 0 + 0 + 8 + 0 + 2 + 1 = (843)_{10}
 \end{array}$$

Q2. Decimal to binary

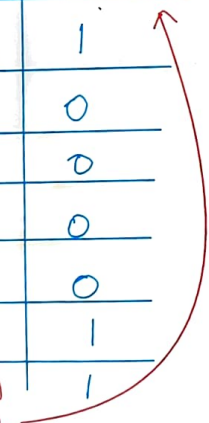
a) $(485)_{10} = (111100101)_2$

2	485	Remainder
2	242	1 LSB
2	121	0
2	60	1
2	30	0
2	15	0
2	7	1
2	3	1
	1	1 MSB



b) $(451)_{10} = (111000011)_2$

2	451	Remainder
2	225	1 LSB
2	112	0
2	56	0
2	28	0
2	14	0
2	7	1
2	3	1
	1	1 MSB



Q3.) Largest 10 bit number

2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
1	1	1	1	1	1	1	1	1	1

Geometric series:

$$a + ar + ar^2 + ar^3 + \dots + ar^{n-1}$$

a = first term

r = common ratio

$$Sum = \frac{a \{ r^n - 1 \}}{r - 1}$$

$$S = \frac{1 \{ 2^{10} - 1 \}}{2 - 1} = 1023$$

assumption: unsigned binary number

Q4. Add the following

a) $(101.11)_2 + (11.01)_2$

$$\begin{array}{r}
 \textcircled{1} \textcircled{1} \textcircled{1} \quad \textcircled{1} \\
 101.11 \\
 + 011.01 \\
 \hline
 1001.00
 \end{array}$$

$$\begin{array}{r}
 2^2 \ 2^1 \ 2^0 \\
 101 \\
 \hline
 4+0+1=5
 \end{array}
 \quad
 \begin{array}{r}
 2^{-1} \ 2^{-2} \\
 11 \\
 \hline
 \frac{1}{2} + \frac{1}{4} = 0.75
 \end{array}$$

$(101.11)_2 = (5.75)_{10}$

$$5.75 + 3.25 = 9.00$$

$$\begin{array}{r} 2^3 \ 2^2 \ 2^1 \ 2^0 \\ 1 \ 0 \ 0 \ 1 \\ \hline 8 + 0 + 0 + 1 = 9 \quad (\text{Verified}) \end{array}$$

$$\begin{array}{r} 2^2 \ 2^1 \ 2^0 \\ 0 \ 1 \ 1 \\ \hline 2 + 1 = 3 \\ (0 \ 1 \ 1 \cdot 0 \ 1)_2 = (3 \cdot 25)_{10} \end{array} \quad \begin{array}{r} 2^{-1} \ 2^{-2} \\ \cdot \ 0 \ 1 \\ \hline \frac{1}{4} = 0.25 \end{array}$$

b) $(10.101)_2 + (1.011)_2$

$$\begin{array}{r} \textcircled{1} \ \textcircled{1} \quad \textcircled{1} \ \textcircled{1} \\ 1 \ 0 \cdot 1 \ 0 \ 1 \\ + \quad 0 \ 1 \cdot 0 \ 1 \ 1 \\ \hline 1 \ 0 \ 0 \cdot 0 \ 0 \ 0 \end{array}$$

$$2.625 + 1.375 = 4$$

$$\begin{array}{r} 2^2 \ 2^1 \ 2^0 \\ 1 \ 0 \ 0 \\ \hline 4 \\ \hline \end{array} \quad (100)_2 = (4)_{10} \quad \text{Verified.}$$

$$\begin{array}{r} 2^1 \ 2^0 \\ 1 \ 0 \\ \hline 2 + 0 = 2 \end{array} \quad \begin{array}{r} 2^{-1} \ 2^{-2} \ 2^{-3} \\ \cdot \ 1 \ 0 \ 1 \\ \hline \frac{1}{2} + \frac{1}{8} = 0.625 \end{array}$$

$$(10.101)_2 = (2.625)_{10}$$

$$\begin{array}{r} 2^0 \\ 1 \\ \hline 1 \end{array} \quad \begin{array}{r} 2^{-1} \ 2^{-2} \ 2^{-3} \\ \cdot \ 0 \ 1 \ 1 \\ \hline \frac{1}{4} + \frac{1}{8} = 0.375 \end{array}$$

$$(10.101)_2 = (1.375)_{10}$$

c) $1000101 \xrightarrow{\quad} 69$

$+ \textcircled{1} 1011001 \xrightarrow{\quad} + 89$

$$\hline 10011110 \xrightarrow{\quad} 158$$

Verified

Q5.) -8 2's complement

$$(8)_{10} = (1000)_2$$

0	0	0	0	1	0	0	0
---	---	---	---	---	---	---	---

↓ 1's

1	1	1	1	0	1	1	1
---	---	---	---	---	---	---	---

$$\begin{array}{r}
 + \\
 \hline
 11111000 \\
 \hline
 \end{array}$$

Q6. 2's complement of 10001 5 bit no.

1	0	0	0	1
---	---	---	---	---

↓ 1's

0	1	1	1	0
---	---	---	---	---

$$\begin{array}{r}
 + \\
 \hline
 01111 \\
 \hline
 \end{array}
 \rightarrow -15$$

Q7.) Solve (5 bit)

a) $10 + 5$

$$(10)_{10} = (01010)_2$$

$$(5)_{10} = (00101)_2$$

$$\begin{array}{r}
 01010 \\
 + 00101 \\
 \hline
 01111 \longrightarrow +15
 \end{array}$$

b) $10 - 5$

$$(10)_{10} = (01010)_2$$

$$(5)_{10} = (00101)_2 \xrightarrow{2's} \begin{array}{r} 11010 \\ + 1 \\ \hline 11011 \end{array}$$

$$\begin{array}{r}
 01010 \\
 + 11011 \\
 \hline
 00101 \\
 \hline
 \end{array}$$

neglect carry on.

+ 5

c) $-10 + 5$

$$(10)_{10} = (01010)_2 \xrightarrow{2's} \begin{array}{r} 10101 \\ + 1 \\ \hline 10110 \end{array}$$

$$(5)_{10} = (00101)_2$$

$$\begin{array}{r}
 10110 \\
 + 00101 \\
 \hline
 11011 \xrightarrow{2's} \begin{array}{r} 00100 \\ + 1 \\ \hline 00101 \end{array} = \boxed{-5}
 \end{array}$$

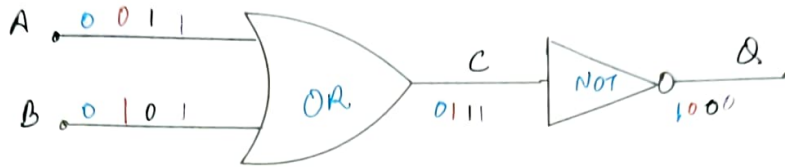
no carry on

d) $-10 - 5$

$$\begin{array}{r}
 10 \xrightarrow{2's} 10110 \quad 5 \xrightarrow{2's} 11011 \\
 \begin{array}{r}
 10110 \\
 + 11011 \\
 \hline
 10001 \xrightarrow{2's} \begin{array}{r} 01110 \\ + 1 \\ \hline 01111 \end{array} \longrightarrow \boxed{-15}
 \end{array}
 \end{array}$$

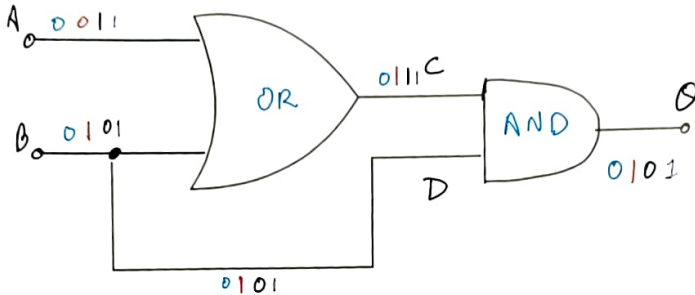
neglect carry

88) a)



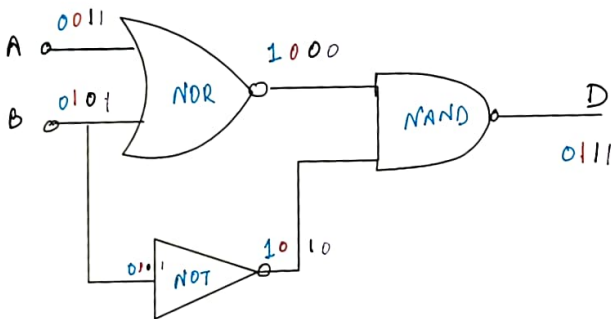
A	B	C	Q
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

b)



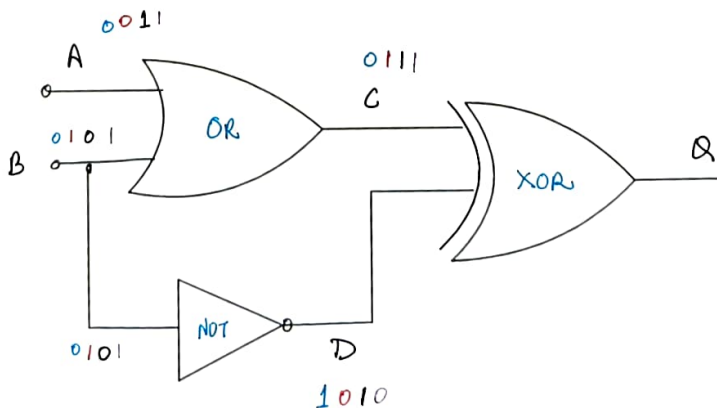
A	B	C	D	Q
0	0	0	0	0
0	1	1	1	1
1	0	1	0	0
1	1	1	1	1

c)



A	B	D
0	0	0
0	1	1
1	0	1
1	1	1

d)

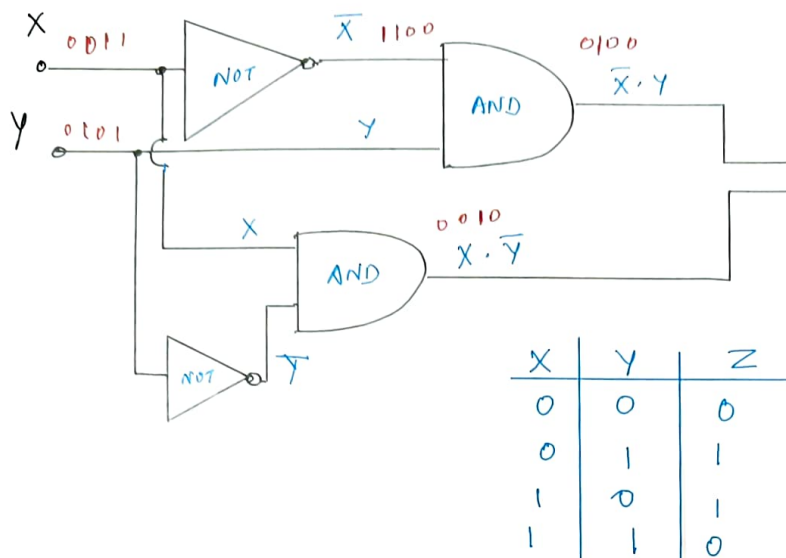


A	B	C	D	Q
0	0	0	1	1
0	1	1	0	1
1	0	1	1	0
1	1	1	0	1

Note:

XOR: $C \oplus D$ gives 0/1 only when inputs are different
 $= C\bar{D} + \bar{C}D$

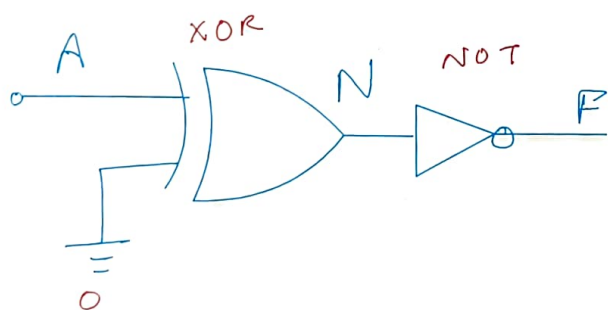
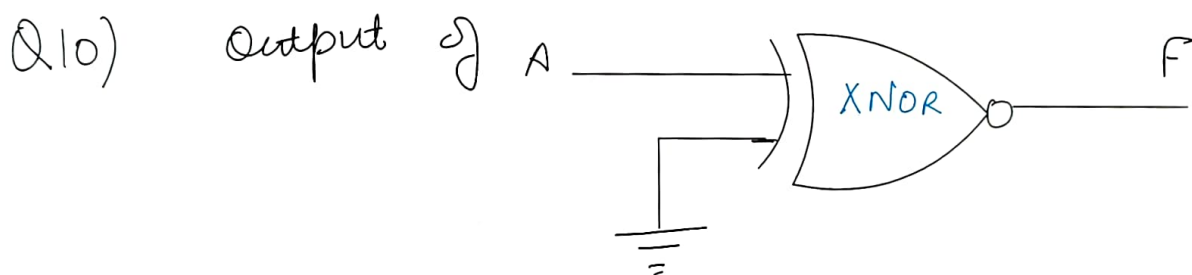
Q9.) In the logic circuit shown below X and Y are digital inputs, Z is a digital output. The equivalent circuit is a



- a) NAND gate
- b) NOR gate
- ☒ c) XOR gate
- d) XNOR gate

$Z = 0110$
 $\bar{X} \cdot \bar{Y} + X \cdot \bar{Y}$
 XOR

$X \oplus Y = Z$



A	0	1	F
0	0	1	1
1	0	0	0

$= \bar{A}$

option (d)